Introduction to R programming language

André Barros

Advanced Data Analysis (ADA) Unit

12th January 2026

Course Outline

- **Day 1**
  - R Basics – Operations, Packages & Classes
  - Loops & Conditions
  - Data Manipulation & Plotting
  - Functions in R – From use to development

- **Day 2**
  - Best Practices (Documentation, Reproducibility, …)
  - AI & Coding
  - tidyverse Universe

Programming & Research

- **Transparency**
- **Reproducibility**
- **Up-to-date methodology**
- **Big Data Era**

Programming & Researcher

- **Documentation**
  - Citations!
  - Job market!

- **Capability & Flexibility**

- **Understanding Data Analysis**

Why R

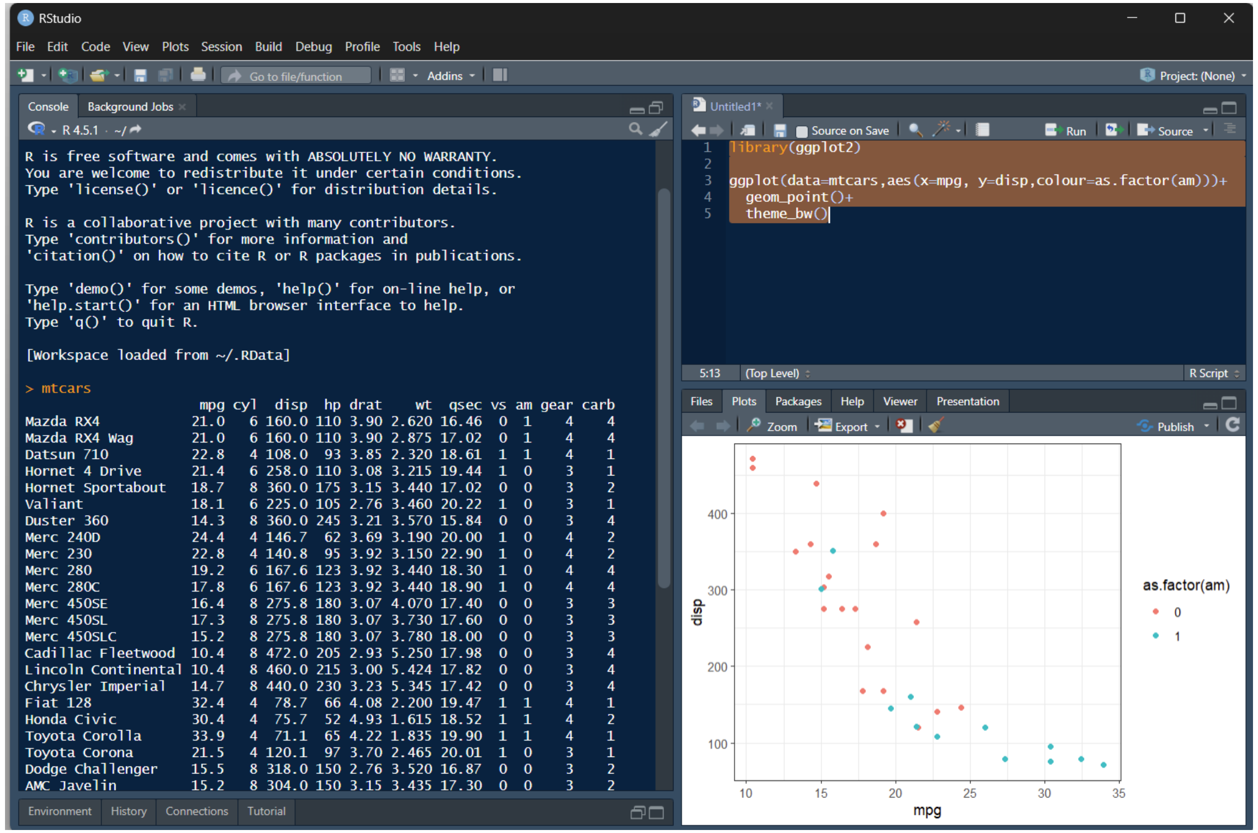
- Statistical Analysis & Data Visualization
- Free-to-use, Open Source
- Thousands of curated & up-to-date packages

Integrated Development Environment (IDE)

RStudio®

Visual Studio Code

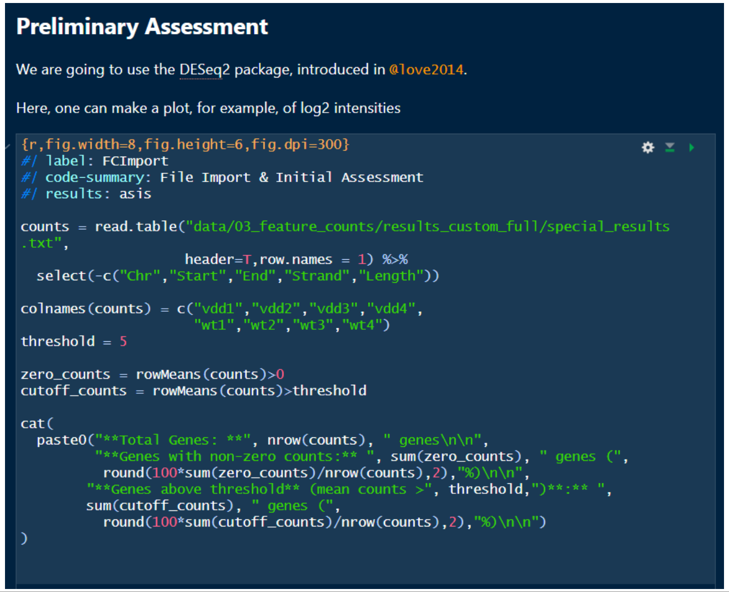
jupyter



Why R

- Statistical Analysis & Data Visualization
- Free-to-use, Open Source
- Thousands of curated & up-to-date packages

Literate Programming

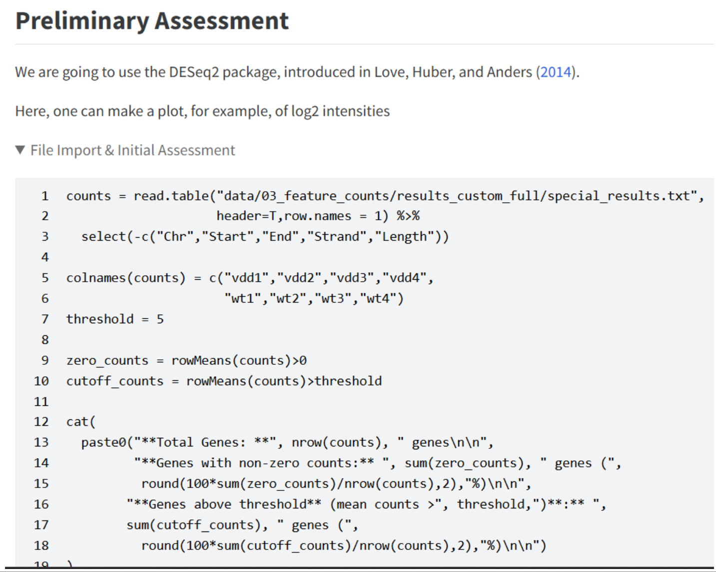


Preliminary Assessment

We are going to use the DESeq2 package, introduced in @love2014.

Here, one can make a plot, for example, of log2 intensities

```
{r,fig.width=8,fig.height=6,fig.dpi=300}
#| label: FCImport
#| code-summary: File Import & Initial Assessment
#| results: asis

counts = read.table("data/03_feature_counts/results_custom_full/special_results
.txt",
                    header=T,row.names = 1) %>%
  select(-c("Chr","Start","End","Strand","Length"))

colnames(counts) = c("vdd1","vdd2","vdd3","vdd4",
                     "wt1","wt2","wt3","wt4")
threshold = 5

zero_counts = rowMeans(counts)>0
cutoff_counts = rowMeans(counts)>threshold

cat(
  paste0("**Total Genes: **", nrow(counts), " genes\n\n",
         "**Genes with non-zero counts:** ", sum(zero_counts), " genes (",
          round(100*sum(zero_counts)/nrow(counts),2),"%)\n\n",
        "**Genes above threshold** (mean counts >", threshold,")**:** ",
        sum(cutoff_counts), " genes (",
          round(100*sum(cutoff_counts)/nrow(counts),2),"%)\n\n")
)
```



Preliminary Assessment

We are going to use the DESeq2 package, introduced in Love, Huber, and Anders (2014).

Here, one can make a plot, for example, of log2 intensities

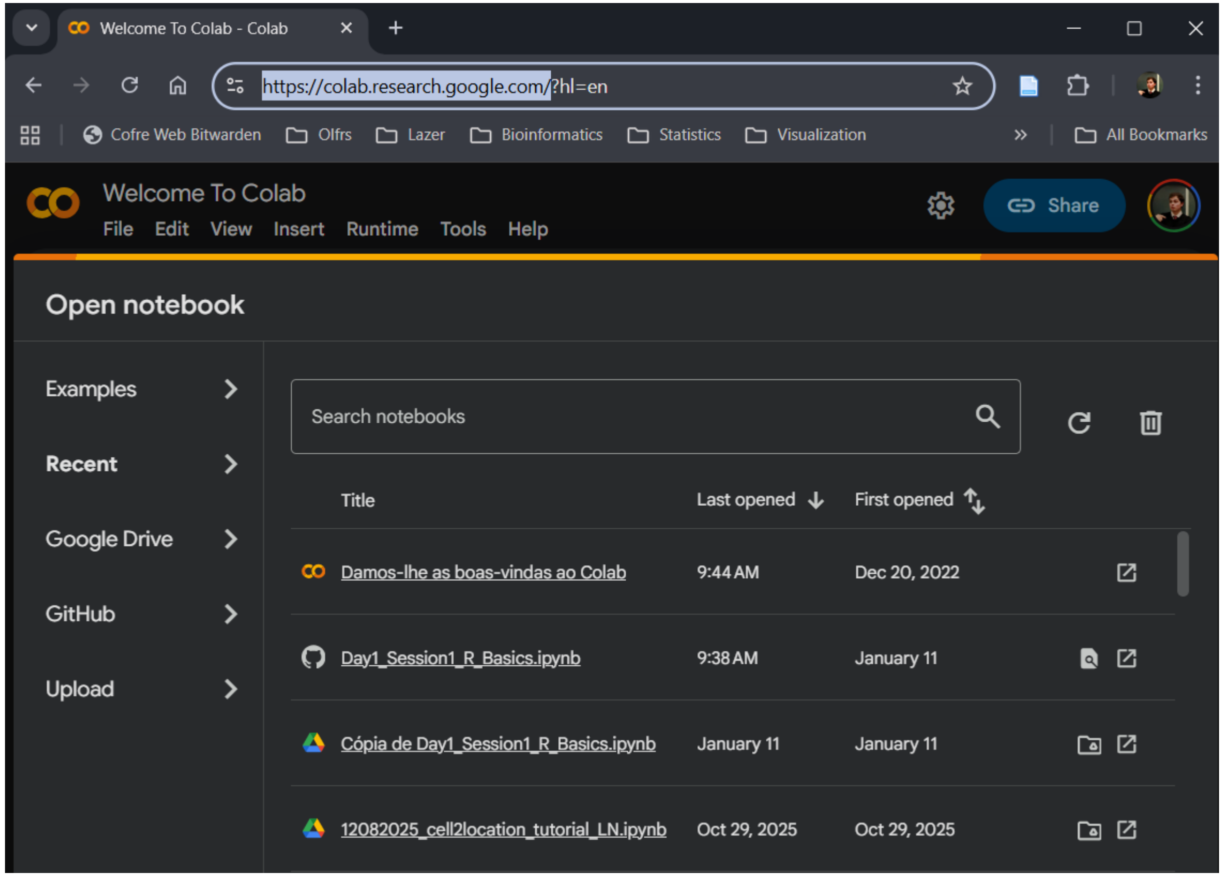
▼ File Import & Initial Assessment

```
counts = read.table("data/03_feature_counts/results_custom_full/special_results.txt",
                    header=T,row.names = 1) %>%
  select(-c("Chr","Start","End","Strand","Length"))

colnames(counts) = c("vdd1","vdd2","vdd3","vdd4",
                     "wt1","wt2","wt3","wt4")
threshold = 5

zero_counts = rowMeans(counts)>0
cutoff_counts = rowMeans(counts)>threshold

cat(
  paste0("**Total Genes: **", nrow(counts), " genes\n\n",
         "**Genes with non-zero counts:** ", sum(zero_counts), " genes (",
          round(100*sum(zero_counts)/nrow(counts),2),"%)\n\n",
        "**Genes above threshold** (mean counts >", threshold,")**:** ",
        sum(cutoff_counts), " genes (",
          round(100*sum(cutoff_counts)/nrow(counts),2),"%)\n\n")
```

Course Material

https://github.com/andrebolerbarros/Intro_R

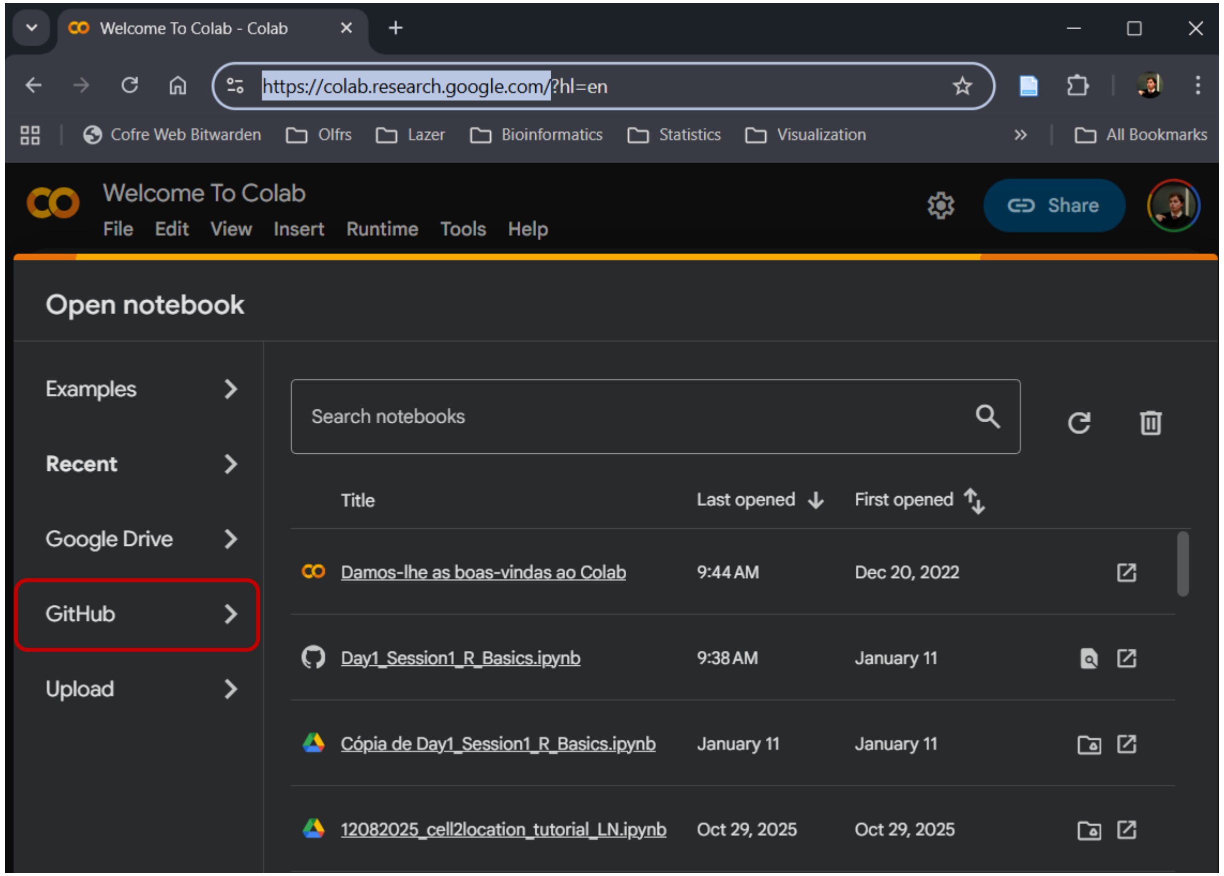
1. Google Colab – colab.research.google.com

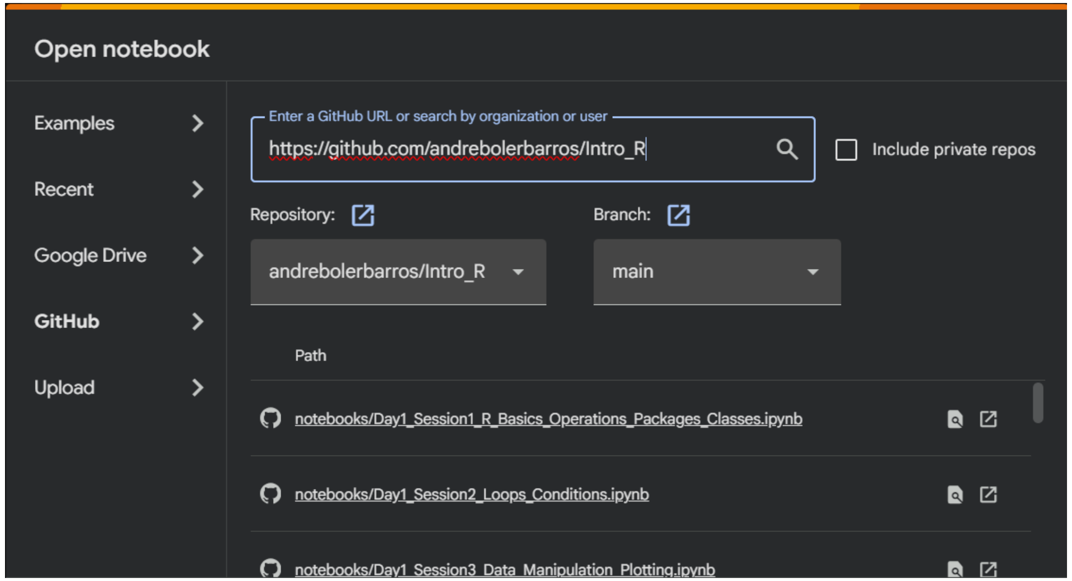


Course Material

https://github.com/andrebolerbarros/Intro_R

1. Google Colab – colab.research.google.com

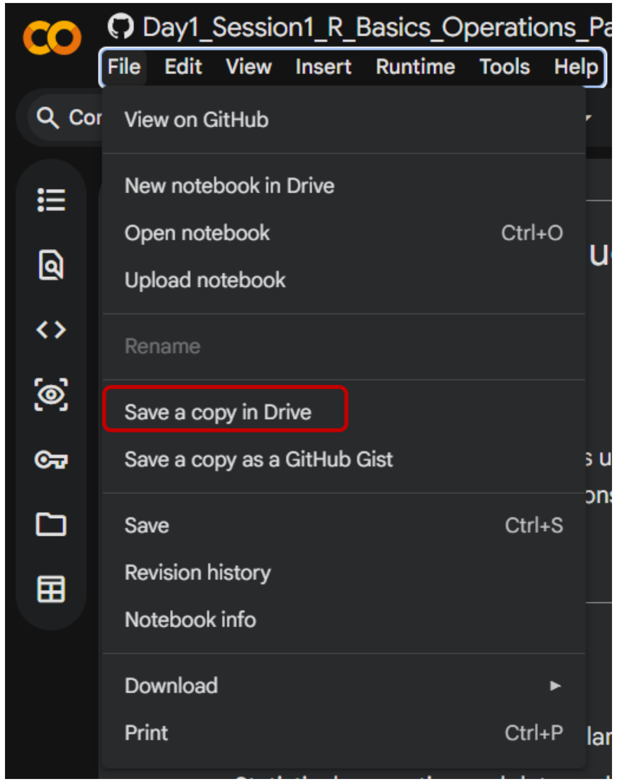


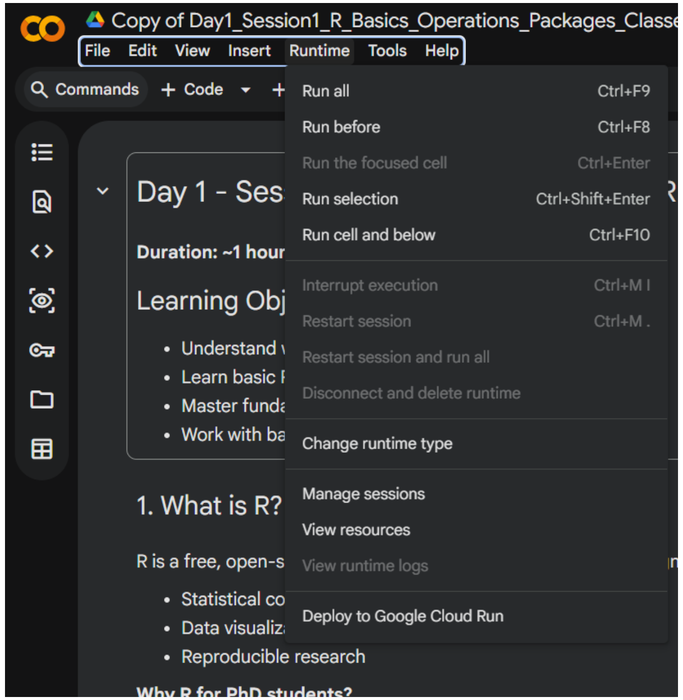


Course Material

https://github.com/andrebolerbarros/Intro_R

2. Save your own copy





Course Material

https://github.com/andrebolerbarros/Intro_R

3. Change kernel (language)